Four acute faces

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Abstract

Four points chosen independently and uniformly on a sphere span a tetrahedron. All four of its faces are acute triangles with probability 1/12, although each face on its own is acute with probability 1/2. We give a short, self-contained proof. Replacing points by their antipodes produces eight tetrahedra, and an obtuse angle of one of them is a short vector sum belonging to another. The short sums fill a staircase in a 4×4 grid, and counting what the staircase can look like reduces the problem to two integrals that Archimedes' hat-box theorem makes easy. The method first appeared in a publicly shared ChatGPT conversation. Every step has been checked in Lean.

1 The question

Choose four points independently and uniformly at random on the unit sphere. They span a tetrahedron. What is the probability that all four faces are acute triangles? Here acute refers to the angles of each triangular face, not to the angles between faces.

A natural first guess goes through one face. Three random points on a sphere form an acute triangle with probability 1/2 [1, Lemma 4]. If the four faces behaved independently, the answer would be $(1/2)^4 = 1/16$. They do not, since any two faces share an edge, and the true answer is larger.

Theorem 1. *Four independent uniform points on the unit sphere in $\mathbb{R}^3$ span a tetrahedron whose four faces are all acute with probability exactly* 1/12*.*

The question was asked on MathOverflow and Mathematics Stack Exchange [3, 4], where simulations suggested 1/12. A proof first appeared in ChatGPT-generated responses in a conversation that Timothy Chow shared publicly and linked from MathOverflow on 14 August 2026 [2]. The method below, antipodal flips, the threshold of Lemma 4 and the count of Section 5, is the method of that conversation, and no priority is claimed for it. What this article offers is an exposition: a complete proof that needs only dot products, a little counting and two one-line integrals.

The plan is short. Section 2 turns each angle condition into a statement about the length of a vector sum. Section 3 flips points to their antipodes and arranges the resulting sums in a grid. Section 4 shows that the short sums form a small staircase, Section 5 counts the possible staircases, and Section 6 averages.

2 Obtuse angles are short sums

Throughout, points of the sphere are unit vectors, and a vector is *short* if its length is less than 1.

Lemma 1. *For unit vectors x, y, z,*

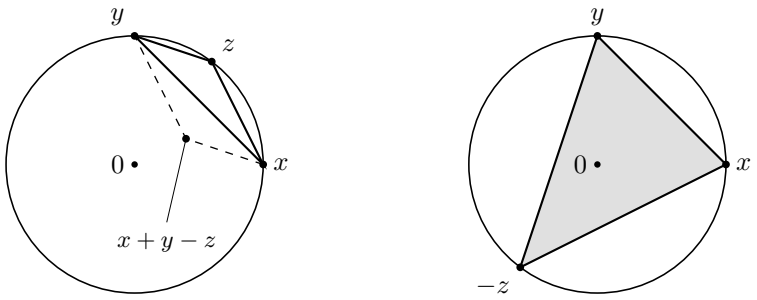
$$(x - z) \cdot (y - z) = \tfrac{1}{2}\big(\|x + y - z\|^2 - 1\big).$$

So the angle at z in the triangle xyz is obtuse exactly when $x + y - z$ is short.

Proof. Using $x \cdot x = y \cdot y = z \cdot z = 1$, both sides equal $x \cdot y - x \cdot z - y \cdot z + 1$. □

The vector $x + y - z$ is the fourth corner of the parallelogram built on the two sides of the triangle at z. Figure 1 shows an example in a plane through the centre: $x = (1, 0, 0)$, $y = (0, 1, 0)$ and $z = \frac{1}{5}(3, 4, 0)$. The angle at z is obtuse, since $(x - z) \cdot (y - z) = -\frac{2}{5}$, and accordingly $\|x + y - z\|^2 = \frac{1}{5}$.

The same vector $x + y - z$ is also the sum of the three vertices of the triangle with vertices x, y and $-z$. In the example that triangle is acute: it contains the centre of its circumscribed circle. So one vector plays two roles. It tests an angle of one triangle, and it is the vertex sum of another. The proof keeps track of these two roles.



angle at z obtuse;
corner $x + y - z$
inside the circle

triangle $x, y, -z$ acute;
its vertex sum is $x + y - z$

Figure 1: Lemma 1 in a plane through the centre. Left: the parallelogram on the sides at z has its fourth corner inside the unit circle, so the angle at z is obtuse. Right: replacing z by $-z$ gives an acute triangle whose vertex sum is the same short vector.

3 Eight tetrahedra in a grid

Let a, b, c, d be the four random points. For every choice of signs $\sigma = (\sigma_1, \sigma_2, \sigma_3, \sigma_4) \in \{\pm 1\}^4$, the points $\sigma_1 a, \sigma_2 b, \sigma_3 c, \sigma_4 d$ span a tetrahedron T_σ. Replacing a uniform point by its antipode leaves it uniform, so every T_σ has the same distribution as $T_{(1,1,1,1)}$. The tetrahedra T_σ and $T_{-\sigma}$ are reflections of each other through the centre and have the same angles, so we count them once. This leaves eight tetrahedra. Let N be the number of them with four acute faces. Each has four acute faces with probability p, the probability we want, so

$$\mathbb{E}N = 8p. \tag{1}$$

The eight tetrahedra are far from independent, but (1) needs only linearity of expectation.

Call a tetrahedron *even* or *odd* according to the parity of the number of minus signs in σ. This is unchanged when σ is replaced by $-\sigma$, because there are four signs, so there are four even and four odd tetrahedra.

A *triangle* here means three of the points $\pm a, \pm b, \pm c, \pm d$ taken from three different pairs, again counted once together with its reflection. Three of the four pairs can be chosen in four ways, and signs on the three chosen pairs in four ways up to reflection, so there are sixteen triangles. Each has a vertex sum, defined up to sign, and so it is either short or not.

Lemma 2. *Every even tetrahedron shares exactly one face with every odd tetrahedron, and every triangle arises in this way from exactly one even and one odd tetrahedron. Two tetrahedra of the same parity share no face.*

Proof. A triangle becomes a tetrahedron when a sign is chosen for the point of the missing pair. The two choices differ in one sign, so one is even and one is odd. Conversely, sign vectors σ of an even and τ of an odd tetrahedron differ in one or three places. In the second case replace τ by

$-\tau$, which differs from σ in one place. The three places where they agree give the common face, and there is no other. Two different tetrahedra of the same parity differ in two places, even after reflection, and share no face. □

So the sixteen triangles fill a 4×4 grid. Put the even tetrahedra on the rows and the odd ones on the columns, and put in each cell the common face of its row and its column. A tetrahedron's own four faces are the four cells of its row, or of its column.

Lemma 3. *A tetrahedron has four acute faces exactly when every short triangle is one of its faces, that is, when every short cell lies in its row or column.*

Proof. Take a face xyz of the tetrahedron. By Lemma 1 its angle at z is obtuse exactly when the triangle $x, y, -z$ is short, and similarly for the angles at x and y. These three triangles and the face xyz are the four triangles on the same three pairs, and the three are not faces of the tetrahedron. Running over the four faces, the twelve angles of the tetrahedron correspond one to one to the twelve triangles that are not its faces. All twelve angles are acute exactly when none of those twelve triangles is short. □

Thus N counts the rows and columns of the grid that contain every short cell.

4 The short cells form a staircase

Which cells are short? Each cell belongs to one row and one column, so it is natural to look at the two tetrahedra involved. For a tetrahedron T write S_T for the sum of its four signed vertices and $w_T = \|S_T\|^2$.

Take an even tetrahedron E and an odd one O. They share a face, with vertex sum L, and their fourth vertices are opposite points v and $-v$. So, up to sign, $S_E = L + v$ and $S_O = L - v$. The cell's sum L is the midpoint of the two totals, and the totals differ by $2v$, a vector of length 2. The parallelogram law, $\|L + v\|^2 + \|L - v\|^2 = 2\|L\|^2 + 2\|v\|^2$, turns this into a statement about lengths.

Lemma 4 (threshold)**.** *For an even tetrahedron E and an odd tetrahedron O, $w_E + w_O = 2\|L\|^2 + 2$, where L is the sum of their common face. In particular their cell is short exactly when $w_E + w_O < 4$.*

So eight numbers, one for each tetrahedron, decide the whole grid.

Order the rows by increasing w_E and the columns by increasing w_O. If a cell is short, then so is every cell above it in its column and every cell to its left in its row, because moving up or left can only decrease $w_E + w_O$. The short cells therefore form a staircase in the top left corner of the grid, a Young diagram. Three more facts fix its size.

Lemma 5. (i) *Among the four triangles on any three pairs, at most one is short. Hence at most four cells are short.* (ii) *No tetrahedron has four short faces. Hence no row or column is entirely short.* (iii) *At least one cell is short.*

Proof. (i) Two of these triangles differ, after reflecting one of them, in the sign of a single vertex v. Their sums are $u + v$ and $u - v$ for some u, and $\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2 \geq 2$, so they are not both short.

(ii) If a tetrahedron has vertices $x_1, \ldots, x_4$ with sum S, its faces have sums $S - x_i$, and $\sum_i \|S - x_i\|^2 = 4\|S\|^2 - 2\|S\|^2 + 4 \geq 4$.

(iii) Here, and only here, we use that the space has dimension 3. Take a tetrahedron with vertices $x_1, \ldots, x_4$ and sum S. Its face opposite x_i has sum $S - x_i$, and

$$\|S - x_i\|^2 = \|S\|^2 - 2\, x_i \cdot S + 1,$$

so that face is short when $x_i \cdot S > \frac{1}{2}\|S\|^2$. Since $S = \sum_i x_i$, the four dot products $x_i \cdot S$ add up to $\|S\|^2$, so their average is only $\frac{1}{4}\|S\|^2$. We need a tetrahedron whose total can be written with less weight on its vertices. The four points satisfy a nontrivial linear relation, and this supplies one. Replace by its antipode each point whose coefficient is negative, and scale so that the largest coefficient is 1. After relabelling, the vertices $x_1, \ldots, x_4$ of one of the eight tetrahedra satisfy

$$x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \lambda_4 x_4 = 0, \qquad 0 \le \lambda_i \le 1.$$

Taking lengths, $1 = \|x_1\| \le \lambda_2 + \lambda_3 + \lambda_4$. Substituting for x_1, the sum of the vertices is

$$S = c_2 x_2 + c_3 x_3 + c_4 x_4, \qquad c_i = 1 - \lambda_i \ge 0, \qquad c_2 + c_3 + c_4 \le 2.$$

If $S = 0$, every face sum $S - x_i$ has length exactly 1. Otherwise some c_i is positive, and if every $x_i \cdot S$ were less than $\frac{1}{2}\|S\|^2$ we would have $\|S\|^2 = \sum_i c_i\,(x_i \cdot S) < \frac{1}{2}\|S\|^2 \sum_i c_i \le \|S\|^2$, which is absurd. So some vertex has $x_i \cdot S \ge \frac{1}{2}\|S\|^2$, and the face opposite it has $\|S - x_i\|^2 \le 1$. Lengths exactly equal to 1 occur with probability zero (Section 6), so this face is short. □

The four orthonormal vectors of $\mathbb{R}^4$ show that (iii) needs the dimension. Every triangle sum then has squared length 3, no cell is short, and all eight tetrahedra are regular.

5 Counting

By Lemma 5(iii) the top left cell is short, so only the first row and the first column can contain every short cell. The first row does so exactly when the staircase is a single row, and the first column exactly when it is a single column. By Lemma 5(i) and (ii), the staircase has between one and four cells and no line of four. Up to interchanging rows and columns, which changes nothing below, there are six possibilities, listed in Figure 2.

We cannot average N directly, but we can average anything built from single cells and pairs of cells in one line. A single cell is short when one point falls in a cap, and two cells in one row are two faces of one tetrahedron, sharing two vertices, which will again need only one cap (Section 6). So let m be the number of short cells and J the number of pairs of short cells in a common row or column, and look for a formula $N = A + Bm + CJ$. The first three shapes in Figure 2 give $A + B = 2$, $A + 2B + C = 1$ and $A + 3B + 3C = 1$, so $A = 4$, $B = -2$ and $C = 1$. The other three shapes fit the same formula, so in every case

$$N = 4 - 2m + J. \tag{2}$$

staircase						
m	1	2	3	3	4	4
J	0	1	3	2	4	4
N	2	1	1	0	0	0
$4 - 2m + J$	2	1	1	0	0	0

Figure 2: The possible staircases of short cells, up to transposition. A single cell lies in one row and one column, which both contain every short cell, so $N = 2$. A row of two or three cells gives $N = 1$. Every other shape spreads over two rows and two columns and gives $N = 0$.

6 Averaging

Each cell is a triangle, and all sixteen triangles have the same distribution, so each is short with the same probability α. A pair of cells in one row consists of two faces of one tetrahedron, which share two vertices, like $a+b+c$ and $a+b+d$. All such pairs have the same distribution, so each is short with the same probability γ. There are 8 lines with $\binom{4}{2} = 6$ pairs each, so taking expectations in (2) gives

$$\mathbb{E}N = 4 - 2 \cdot 16\alpha + 48\gamma. \tag{3}$$

Both probabilities come from one fact, Archimedes' hat-box theorem: the height of a uniform point on the sphere, measured along any fixed axis, is uniform on $[-1, 1]$. Equivalently, a band between two parallel planes at distance h apart has area $2\pi h$, the same as the matching band on the circumscribed cylinder. (At height h the circle of latitude has radius $r = \sqrt{1-h^2}$. A thin band there has circumference $2\pi r$ and sloping width dh/r, so its area is $2\pi\, dh$ whatever the height.)

Fix a and b, and let $s = \|a+b\|$ and $u = (a+b)/s$. Then

$$\|a+b+c\|^2 = s^2 + 2s\,(c \cdot u) + 1,$$

which is less than 1 exactly when the height $c \cdot u$ of c along u is below $-s/2$. That is a cap of height $1 - s/2$, so

$$\Pr(a+b+c \text{ short} \mid a, b) = \frac{1 - s/2}{2} = \frac{1-t}{2}, \qquad t = \frac{s}{2}.$$

Given a and b, the points c and d are independent and fall in the same cap, so both $a+b+c$ and $a+b+d$ are short with probability $\big((1-t)/2\big)^2$ (Figure 3). The cap is random, so this square must be taken before averaging.

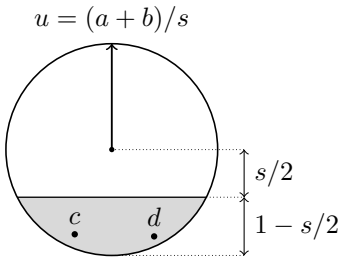


Figure 3: With a and b fixed, $a+b+c$ is short exactly when c lies in the shaded cap, whose height is $1 - s/2$. By Archimedes the cap has the fraction $(1 - s/2)/2$ of the sphere's area.

It remains to average over a and b. Since $s^2 = 2 + 2\,a \cdot b$ and $a \cdot b$ is uniform on $[-1, 1]$, again by Archimedes, $t^2 = s^2/4$ is uniform on $[0, 1]$. Hence $\mathbb{E}t = \int_0^1 \sqrt{v}\, dv = \frac{2}{3}$ and $\mathbb{E}t^2 = \frac{1}{2}$, and

$$\alpha = \frac{1 - \mathbb{E}t}{2} = \frac{1}{6}, \qquad \gamma = \frac{1 - 2\,\mathbb{E}t + \mathbb{E}t^2}{4} = \frac{1 - \frac{4}{3} + \frac{1}{2}}{4} = \frac{1}{24}.$$

This agrees with the known result for a single face. By Lemma 1, a given angle of a random triangle on the sphere is obtuse with probability $\alpha = \frac{1}{6}$. At most one angle of a triangle can be obtuse, so the triangle is acute with probability $1 - 3 \cdot \frac{1}{6} = \frac{1}{2}$.

Substituting into (3),

$$\mathbb{E}N = 4 - \frac{32}{6} + \frac{48}{24} = \frac{2}{3}, \qquad \text{so} \qquad p = \frac{\mathbb{E}N}{8} = \frac{1}{12}$$

by (1). This proves Theorem 1.

Two details were set aside. Given a and b with $s > 0$, the squared length $\|a + b + c\|^2$ is uniform on an interval, so it equals 1 with probability zero, and $s = 0$ itself has probability zero. The same holds for every one of the sixteen triangles, so with probability one no vertex sum has length exactly 1, no angle is right, and Lemma 4 has no ties at the threshold. Ties among the w_T do not matter, since any ordering consistent with the values gives a staircase.

7 Remarks

For comparison, Wendel's theorem [5] gives the probability that the convex hull of random points contains the centre by the same device of antipodal flips, but there exactly one flip in each family succeeds. Here the number of successes N is 0, 1 or 2, which is why a count such as (2) is needed.

A Lean 4 formalisation of Theorem 1, using Mathlib, accompanies the article as supplementary material. It states the result for the uniform measure on the sphere and proves it, including Archimedes' theorem and both integrals, using only Lean's standard axioms. It follows the plan above, although some steps take different routes: for instance, it proves the count in the equivalent form $N + C = 4$, where $C = 2m - J$ is the number of tetrahedra with one or two short faces, by checking every pattern of short triangles allowed by restrictions of the kind in Lemma 5. Sixteen deliberately falsified variants of its statements fail to compile.

Declaration of AI use

The exposition is my own. I used Claude Opus 5.5 (Anthropic) to help draft the paper and to write the Lean formalisation. The theorem and the method of proof are credited to [2].

References

[1] A. Akopyan, H. Edelsbrunner and A. Nikitenko, The beauty of random polytopes inscribed in the 2-sphere, *Exp. Math.* **34** (2025), 327–341.

[2] T. Chow (shared by), Tetrahedron acute faces proof, ChatGPT-generated responses in a publicly shared conversation, linked from a comment on MathOverflow question 498296 on 14 August 2026, `https://chatgpt.com/share/6a7e8742-01e0-83ea-81de-533ea9dcb597`.

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